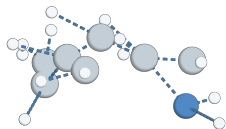


a From composition to 3D structure

Atom identities · total charge · spin multiplicity

Harmonic source



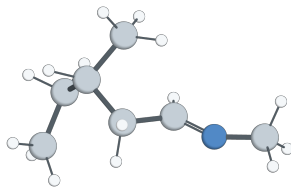
$$X_0 \sim q_0(\cdot \mid c)$$

Auxiliary edges only



LatentSpring flow
EGNN + corrections

Neural sampling



$$X_{\text{gen}} \in \mathbb{R}^{23 \times 3}$$

C7H15N · neutral singlet

b Correct the predicted endpoint

Geometry

$$g = G_\psi(H, p; c)$$

$$H_g = H + (1 - p)g$$



Physics

$$f = A_\phi(X, H_g, p; c)$$

Force-based supervision



Endpoint update

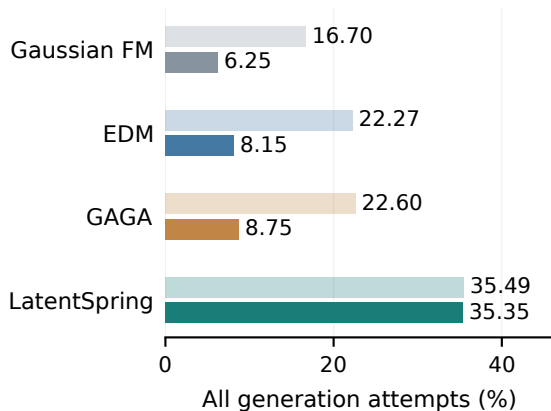
$$\Delta H = (1 - p)(g + \eta f)$$

Flow or diffusion

Training

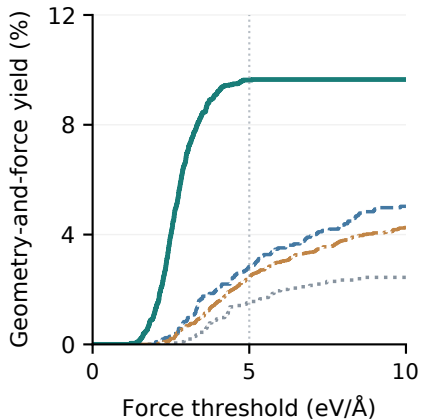
Reference geometry → backbone + geometry head

eSEN force targets → physical head

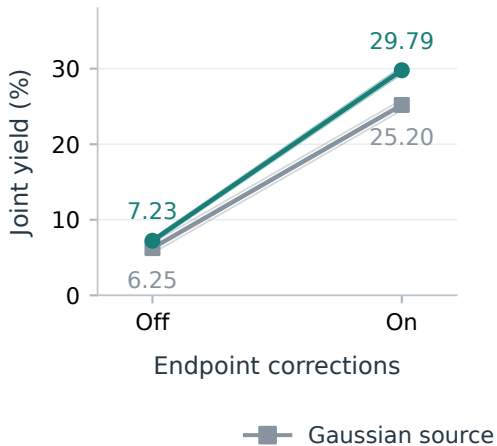
a Graph and force

Graph-valid

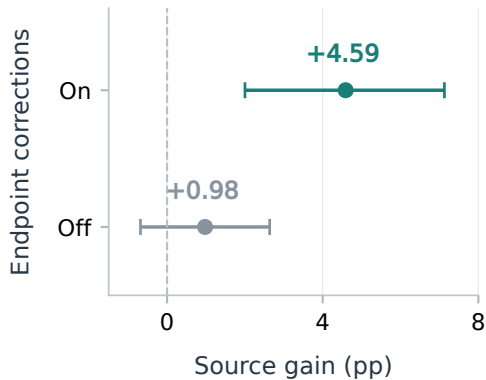
Graph-valid + force ≤ 5 eV/Å

b Local geometry

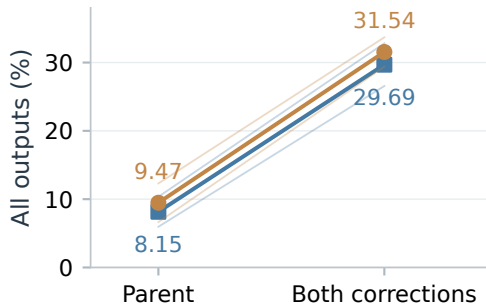
A Four configurations



B Harmonic source gain

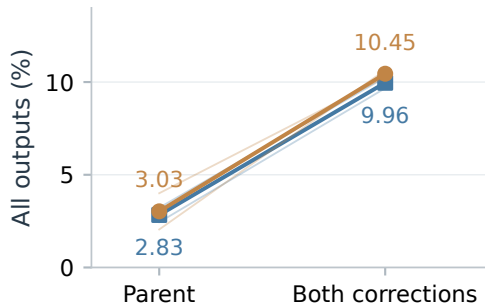


A Graph and force



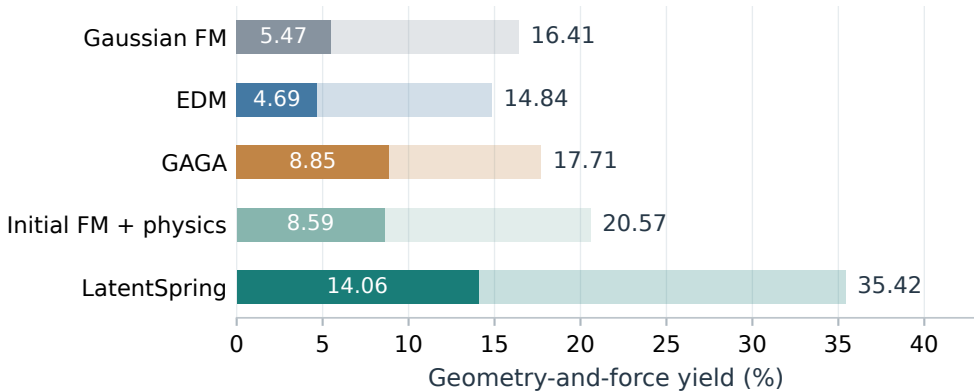
EDM

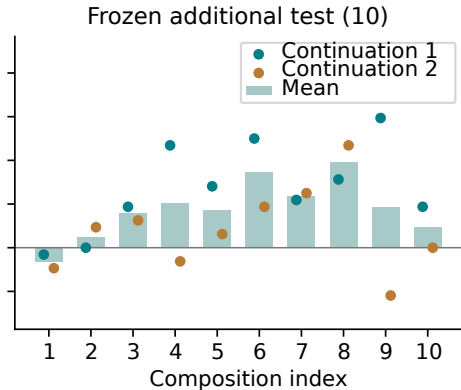
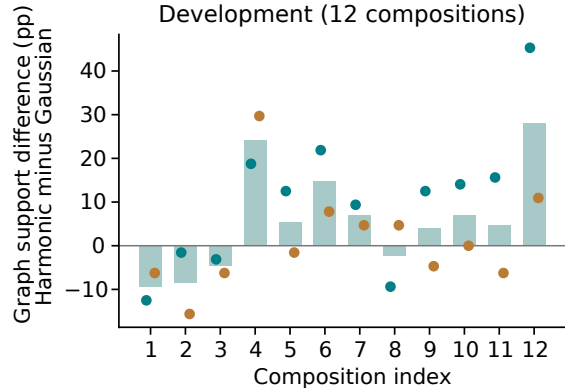
B Geometry and force



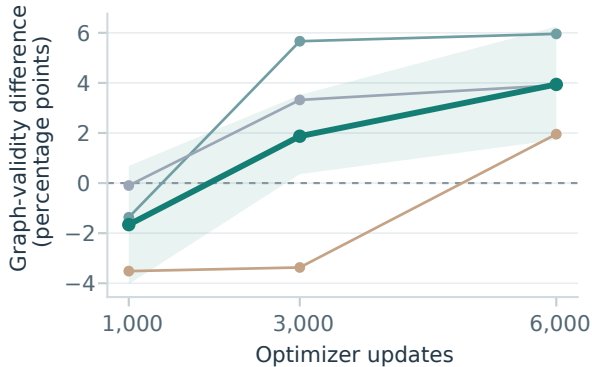
GAGA

Common-element organic compositions

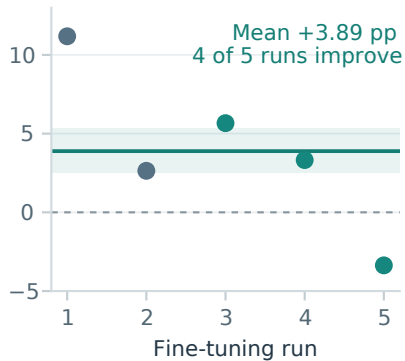




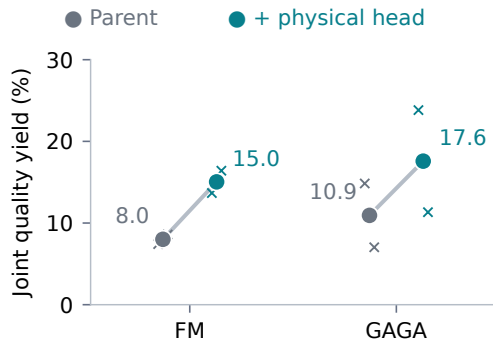
a Training budget



b Five training runs



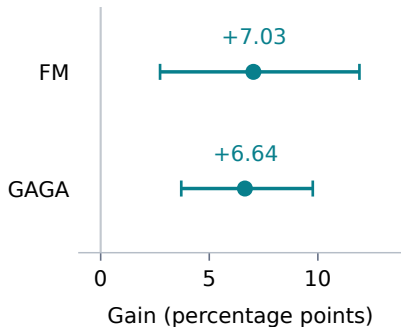
Run 3 Run 4 Run 5 Mean

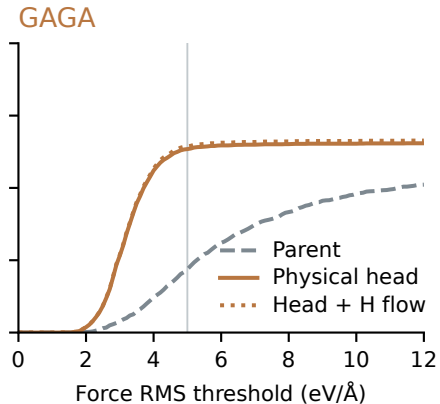
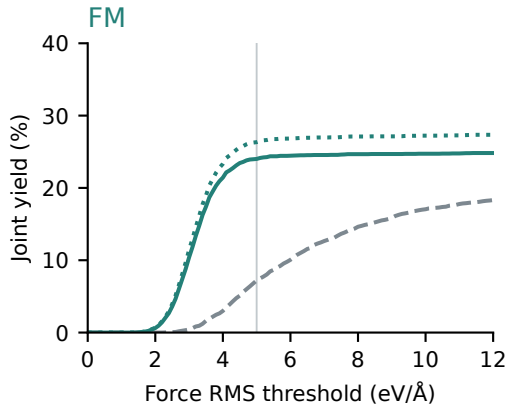
a Same EGNN · new compositions

512 raw outputs per method · no geometry optimization

b Paired improvement

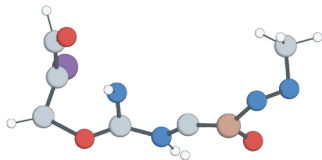
Mean and 95% interval





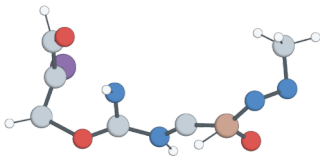
a Hydrogen placement around one fixed heavy skeleton

Parent



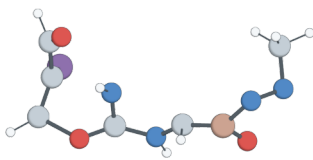
Detached hydrogen

Radial rule



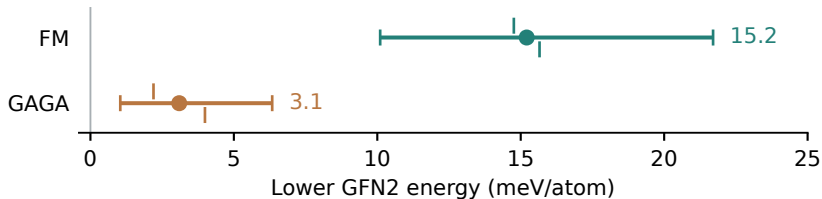
Graph-valid

Conditional flow



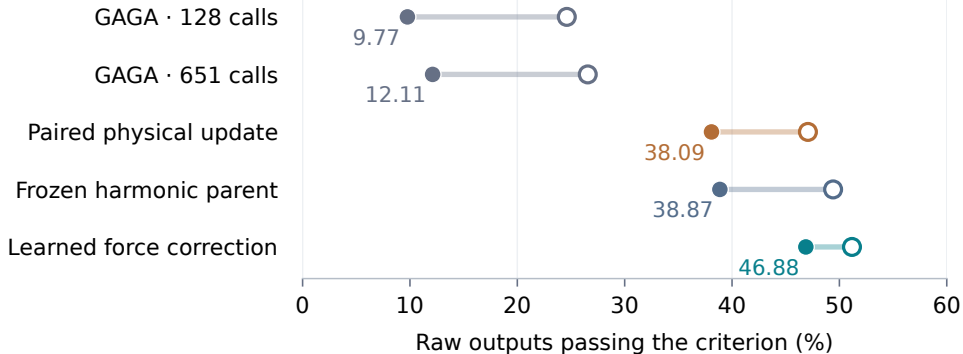
Graph-valid

b Energy reduction relative to the radial rule

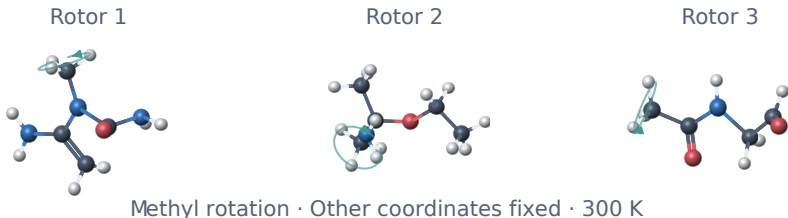


Same compositions · raw generated coordinates

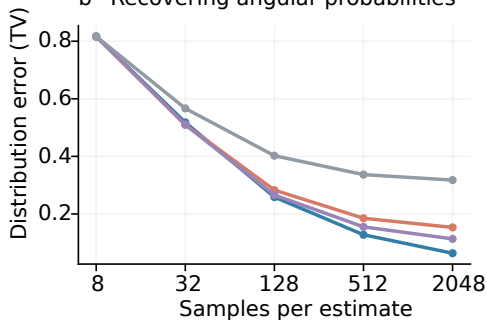
● Graph-valid and force RMS ≤ 5 eV/Å ○ Graph-valid



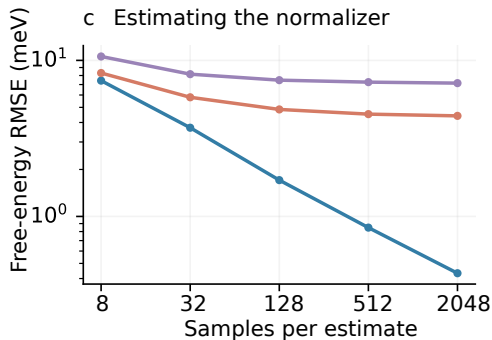
a Controlled methyl rotation on three molecules



b Recovering angular probabilities



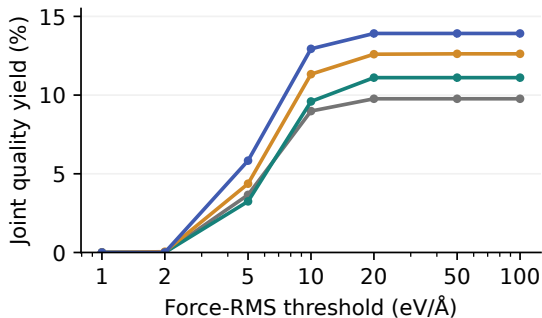
c Estimating the normalizer



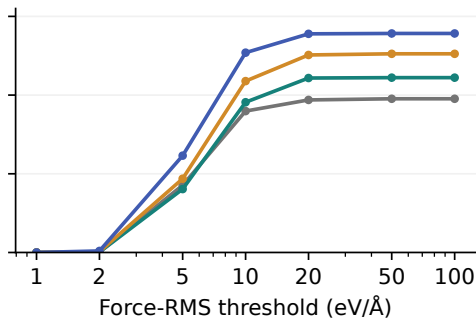
Full work Energy only No Jacobian Unweighted

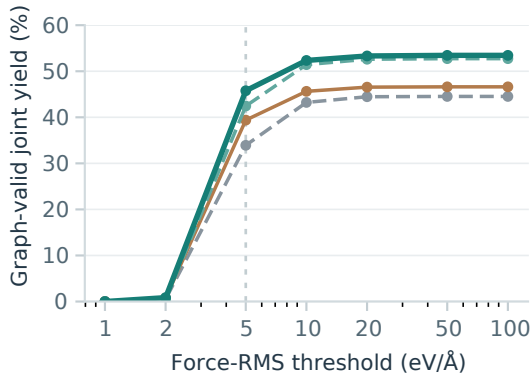
—●— Gaussian FM —●— Harmonic FM —●— EDM —●— GAGA

eSEN



GFN2-xTB



a eSEN**b Independent GFN2-xTB**